

Streaming Platform Subscription Costs vs. Film Viewer Enjoyment

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PURPOSE

Streaming prices keep rising, so this project asks the simple question; is paying more for a streaming tier associated with a better movie-watching experience? It links historical subscription prices (2022 - 2026) to my own personal film viewing log to test whether cost and enjoyment are related.

DATA and METHODS

- **Pricing Data:** A timeline of monthly subscription costs by platform from 2022 - 2026
- **Viewing Data:** A personal log of 478 films, each with a rating (1-10), viewing date, and platform streamed from
- **Join:** Each film was matched to the price in effect on its viewing date using a date-range (inequality) join
- **Tests:** Pearson correlation (Rating vs. Price), One-Way ANOVA (Rating by Platform), and Tukey HSD Post-Hoc Comparisons

KEY FINDINGS

TEST	RESULT	TAKEAWAY
Pearson Correlation	$r = 0.1351$ $p = 0.003$	Statistically Significant but Weak Positive Association
Coefficient of Determination	$r^2 = 0.018$	Price Accounts for only 1.8% of Rating Variance
One-Way ANOVA	$p = 0.00127$	Average Ratings differ across Platforms
Tukey HSD	HBO Max vs. Disney+ +1.66 points $p_{adj} = 0.0002$	The Only Significant Pairwise Gap

CONCLUSION

Higher subscription cost is associated with slightly higher film ratings, but the relationship is practically negligible. With 478 films, even a small effect reaches statistical significance, yet price explains less than 2% of the variation in ratings. What appears to matter more is the streaming service itself; films watched on HBO Max were rated 1.66 points higher on average (on a 10-point scale) than those watched on Disney+ (no other streaming service pair differed significantly). For budget-conscious viewers, price alone is a poor guide for value, whereas catalog fit likely matters more.

LIMITATIONS

- Data comes from one viewer (my own), so results reflect personal taste and may not generalize to wider public
- Price and platform are intertwined, so the correlation may reflect platform differences rather than a separate "price effect"
- Unmeasured factors such as genre, director, and mood account for most of the variance
- Non-significant platform pairs show no *detected* difference, not proof of equal enjoyment

NEXT STEPS

- Add genre and release year as controls in a regression model
- Check ANOVA assumptions (normality, equal variances) & Kruskal-Wallis test as a robustness check
- Add cost per film watched as a second value metric